

Lemma: G 是一个任意的群, 因此 G 是非空集合. 定义 G 在 G 上的群作用 $\alpha: G \times G \rightarrow G$ 则有:

$$(g, x) \mapsto gxg^{-1}$$

① 映射 $\alpha: G \times G \rightarrow G$ 是一个群作用, 称为 G 在 G 上的共轭作用

② 对 $\forall x \in G$, 有: $\text{Stab}_G(x) = Z_G(x)$

③ ~~$G^G = Z_G$~~ $G^G = Z_G$

Proof: 对 $\forall x \in G$, 有: $\alpha(1_G, x) = 1_G \times 1_G^{-1} = 1_G \times 1_G = x 1_G = x$

对 $\forall g_1, g_2 \in G, \forall x \in G$, 有: $\because g_1, g_2 \in G \therefore g_1 g_2 \in G$.

$$\alpha(g_1, \alpha(g_2, x)) = \alpha(g_1, g_2 x g_2^{-1}) = g_1 (g_2 x g_2^{-1}) g_1^{-1}$$

$$= (g_1 g_2) x (g_2^{-1} g_1^{-1}) = (g_1 g_2) x (g_1 g_2)^{-1} = \alpha(g_1 g_2, x)$$

$\therefore$ 映射 $\alpha: G \times G \rightarrow G$ 是一个群作用. 称 α 为 G 在 G 上的共轭作用.

$$\text{对 } \forall x \in G, \text{ 有: } \text{Stab}_G(x) = \{g \in G \mid gx = x\} = \{g \in G \mid gxg^{-1} = x\}$$

$$Z_G(x) = \{\alpha \in G \mid x\alpha = \alpha x\}.$$

$$\text{对 } \forall g \in \text{Stab}_G(x), \text{ 有: } gxg^{-1} = x. \quad \therefore gx = xg \quad \therefore xg = gx$$

$$\therefore g \in Z_G(x). \quad \therefore \text{Stab}_G(x) \subseteq Z_G(x)$$

$$\text{对 } \forall \alpha \in Z_G(x), \text{ 有: } x\alpha = \alpha x \quad \therefore \alpha x \alpha^{-1} = x \alpha \alpha^{-1} = x 1_G = x.$$

$$\therefore \alpha \in \text{Stab}_G(x) \quad \therefore Z_G(x) \subseteq \text{Stab}_G(x) \quad \therefore \text{Stab}_G(x) = Z_G(x)$$

$\therefore$ 群 G 作用在非空集合 G 上

$$\begin{aligned}\therefore G^G &= \{x \in G \mid \text{对 } \forall g \in G \text{ 都有 } gx = x\} \\ &= \{x \in G \mid \text{对 } \forall g \in G \text{ 都有 } gxg^{-1} = x\} \subseteq G.\end{aligned}$$

$$\therefore Z_G = \{x \in G \mid \text{对 } \forall g \in G \text{ 都有 } xg = gx\} \subseteq G.$$

$$\text{对 } \forall x \in G^G, \text{ 有: 对 } \forall g \in G, \therefore x \in G^G \therefore gxg^{-1} = x$$

$$\therefore gx = xg \quad \therefore xg = gx \quad \therefore x \in Z_G \quad \therefore G^G \subseteq Z_G.$$

$$\text{对 } \forall x \in Z_G, \text{ 有: 对 } \forall g \in G, \therefore x \in Z_G \therefore xg = gx$$

$$\therefore gxg^{-1} = xgg^{-1} = x|_G = x \quad \therefore x \in G^G \quad \therefore Z_G \subseteq G^G$$

$$\therefore G^G = Z_G \quad \square$$

Lemma: G 是群, $x, y \in G$, $xy = |_G$. 则有: $yx = |_G$.

$$\text{Proof: } \therefore x, y \in G \quad \therefore xy \in G, yx \in G$$

$$\therefore \exists (yx)^{-1} \in G, \text{ s.t. } (yx)(yx)^{-1} = |_G = (yx)^{-1}(yx)$$

$$\therefore (yx)(yx) = y(xy)x = y \cdot |_G \cdot x = yx$$

$$\begin{aligned}\therefore |_G &= (yx)(yx)^{-1} = ((yx)(yx)) (yx)^{-1} \\ &= (yx)((yx)(yx)^{-1}) = (yx)|_G = yx\end{aligned}$$

$$\therefore yx = |_G \quad \square$$

定义：群 G 作用在非空集合 X 上，记

$$X^G = \{x \in X \mid \text{对 } \forall g \in G \text{ 都有 } gx = x\} \subseteq X$$

集合 X^G 的元素称为 X 在 G 作用下的不动点。

Lemma：群 G 作用在非空集合 X 上，对 $\forall x \in X$ ，有：

$$x \in X^G \iff Gx = \{x\} \iff |Gx| = 1$$

Proof：对 $\forall x \in X$ ，有： $Gx = \{gx \mid g \in G\} \subseteq X$ 。

$$\because 1_G \in G \text{ 且 } 1_G x = x \quad \therefore x = 1_G x \in Gx \quad \therefore \{x\} \subseteq Gx.$$

$$\therefore |Gx| = 1 \iff Gx = \{x\} \quad \text{右边的“} \iff \text{”得证。}$$

左 $(\Rightarrow)$ ：~~已经证明~~ 已经证明了 $\{x\} \subseteq Gx$ 。

任取 Gx 中的一元： gx (其中 $g \in G$)。 $\because x \in X^G$ ， $g \in G$

$$\therefore gx = x \quad \therefore gx \in \{x\} \quad \therefore Gx \subseteq \{x\} \quad \therefore Gx = \{x\}$$

左 $(\Leftarrow)$ ：已知 $x \in X$ 。

$$\text{对 } \forall g \in G, \quad \because g \in G \quad \therefore gx \in Gx = \{x\} \quad \therefore gx = x$$

$$\therefore x \in X^G \quad \square$$

Lemma：群 G 作用在非空集合 X 上，对 $\forall x \in X$ ，有： $|Gx| \geq 1$

(注意：轨道的基数完全可以是 $+\infty$)

Proof：对 $\forall x \in X$ ，有： $Gx = \{gx \mid g \in G\} \subseteq X$ 。

$$\because 1_G \in G \text{ 且 } 1_G x = x \quad \therefore x = 1_G x \in Gx \quad \therefore \{x\} \subseteq Gx$$

$$\therefore |Gx| \geq 1 \quad \square$$

Lemma (群作用的轨道方程) G 是群, X 是非空有限集合.

群 G 作用在非空有限集合 X 上. 设所有包含大于 1 个元素的轨道

为 $O_1, \dots, O_k$ (其中 $k \in \mathbb{Z}_{\geq 0}$). 对 $\forall i \in \{1, \dots, k\}$, 任取 $x_i \in O_i$.
(因此 $x_i \in X$)

$$\text{则有: } |X| = |X^G| + \sum_{i=1}^k (G : \text{Stab}_G(x_i))$$

Proof: $\because$ 群 G 作用在非空有限集合 X 上.

$\therefore$ 设所有只包含一个元素的轨道为 $T_1, \dots, T_l$ (其中 $l \in \mathbb{Z}_{\geq 0}$)

设 $T_1 = \{y_1\}, \dots, T_l = \{y_l\}$ (其中 $y_1, \dots, y_l \in X$)

$$\therefore T_1 = Gy_1, \dots, T_l = Gy_l$$

$$\text{对 } \forall x \in X, \text{ 有: } x \in X^G \Leftrightarrow |Gx| = 1 \Leftrightarrow x \in \{y_1, \dots, y_l\}$$

$$\therefore X^G = \{y_1, \dots, y_l\}$$

$$\therefore X = T_1 \sqcup \dots \sqcup T_l \sqcup O_1 \sqcup \dots \sqcup O_k$$

$$\therefore |X| = |T_1| + \dots + |T_l| + |O_1| + \dots + |O_k|$$

$$= l + |O_1| + \dots + |O_k| = |X^G| + |O_1| + \dots + |O_k|$$

$$= |X^G| + |Gx_1| + \dots + |Gx_k| = |X^G| + \sum_{i=1}^k |Gx_i|$$

$$= |X^G| + \sum_{i=1}^k |G/\text{Stab}_G(x_i)| = |X^G| + \sum_{i=1}^k (G:\text{Stab}_G(x_i)) \quad \square$$

定义 (p -群) G 是群, p 是素数. 若 $|G| = p^m$ (其中 $m \in \mathbb{Z}_{\geq 0}$), 则称 G 是 p -群.

Lemma: G 是 p -群, G 作用在非空有限集合 X 上, 则有:

$$|X| \equiv |X^G| \pmod{p}$$

Proof: 设所有包含大于 1 个元素的轨道为 $O_1, \dots, O_k$ (其中 $k \in \mathbb{Z}_{\geq 0}$)
 对 $\forall i \in \{1, \dots, k\}$, 任取 $x_i \in O_i$. 则已经证明了

$$|X| = |X^G| + \sum_{i=1}^k (G:\text{Stab}_G(x_i))$$

对 $\forall i \in \{1, \dots, k\}$, $\because x_i \in O_i \subseteq X \therefore x_i \in X \therefore \text{Stab}_G(x_i) \leq G$

$\because G$ 是群, $\text{Stab}_G(x_i) \leq G \therefore |G| = |\text{Stab}_G(x_i)| \cdot (G:\text{Stab}_G(x_i))$

$\because G$ 是 p -群 $\therefore |G| = p^m$ ($m \in \mathbb{Z}_{\geq 0}$), $|G| \in \mathbb{Z}_{\geq 1}$

$\because \text{Stab}_G(x_i) \leq G \therefore 1_G \in \text{Stab}_G(x_i) \therefore \text{Stab}_G(x_i) \neq \emptyset$

$\therefore |\text{Stab}_G(x_i)| \in \mathbb{Z}_{\geq 1} \therefore (G:\text{Stab}_G(x_i))$ 不可能是 0 或 $+\infty$

$\therefore (G:\text{Stab}_G(x_i)) \in \mathbb{Z}_{\geq 1} \therefore \frac{|G|}{(G:\text{Stab}_G(x_i))} = |\text{Stab}_G(x_i)| \in \mathbb{Z}_{\geq 1}$

$\therefore (G:\text{Stab}_G(x_i)) \mid |G| \therefore (G:\text{Stab}_G(x_i)) \mid p^m$

$\therefore (G:\text{Stab}_G(x_i)) = p^{\alpha_i}$ (其中 $\alpha_i \in \mathbb{Z}_{\geq 0}$ 且 $\alpha_i \leq m$)

假设 $\alpha_i = 0$, 则 $(G: \text{Stab}_G(x_i)) = p^{\alpha_i} = p^0 = 1$

$$\therefore |G| = |\text{Stab}_G(x_i)| \quad \because \text{Stab}_G(x_i) \leq G \quad \therefore \text{Stab}_G(x_i) = G$$

$\therefore x_i \in O_i \subseteq X$, O_i 是包含大于 1 个元素的轨道

$$\therefore O_i = Gx_i \text{ 且 } |Gx_i| = |O_i| > 1$$

$$\therefore |_G x_i = x_i \quad \therefore x_i = |_G x_i \in Gx_i \quad \therefore \{x_i\} \subseteq Gx_i$$

任取 Gx_i 中的一个元素 gx_i (其中 $g \in G$) $\therefore g \in G = \text{Stab}_G(x_i)$

$$\therefore gx_i = x_i \quad \therefore gx_i \in \{x_i\} \quad \therefore Gx_i \subseteq \{x_i\} \subseteq Gx_i$$

$$\therefore Gx_i = \{x_i\} \quad \therefore |Gx_i| = |\{x_i\}| = 1 \quad \therefore 1 > 1 \text{ 矛盾}$$

$$\therefore \alpha_i \neq 0 \quad \therefore \alpha_i \in \mathbb{Z}_{\geq 1} \text{ 且 } \alpha_i \leq m$$

$$\therefore |X| - |X^G| = \sum_{i=1}^k (G: \text{Stab}_G(x_i)) = \sum_{i=1}^k p^{\alpha_i}$$

$$\therefore p \mid |X| - |X^G| \quad \therefore |X| \equiv |X^G| \pmod{p} \quad \square$$

Remark: 如果 $k=0$, 则所有轨道的基数都等于 1.

则 $X = X^G$. 引理的结论显然成立.

定义 (群的中心) G 是群, 定义 $Z_G := \{x \in G \mid \text{对 } \forall g \in G, \text{ 都有 } xg = gx\}$.

我们称 Z_G 是群 G 的中心

Lemma: G 是群, 则有 $Z_G \triangleleft G$

Proof: 已经证明了 $Z_G \leq G$. $\therefore G$ 是群, $Z_G \leq G$.

对 $\forall g \in G$, 任取 gZ_gg^{-1} 中的一元: gxg^{-1} (其中 $x \in Z_g$),

$$\because x \in Z_G, g \in G \quad \therefore x \in G \text{ 且 } xg = gx$$

$$\therefore g \times g^{-1} = (xg)g^{-1} = x(gg^{-1}) = x/1 = x \in Z_G$$

$$\therefore gZ_Gg^{-1} \subseteq Z_G \quad \therefore Z_G \triangleleft G \quad \square$$

Lemma: G 是非平凡的 p -群, 则 $Z_G \neq \{1_G\}$, $\{1_G\} \subsetneq Z_G$.

Proof: $\because G$ 是 p -群 $\therefore |G| = p^m$ (其中 p 是素数, $m \in \mathbb{Z}_{\geq 0}$).

$\because G$ 是非平凡的 p -群 $\therefore m \neq 0 \quad \therefore m \in \mathbb{Z}_{\geq 1}$

$\therefore |G| = p^m$ (其中 p 是素数, $m \in \mathbb{Z}_{\geq 1}$)

$\because 1_G \in G$, 且对 $\forall g \in G$, 都有 $1_G \cdot g = g = g \cdot 1_G \quad \therefore 1_G \in Z_G$

$\therefore \{1_G\} \subseteq Z_G$

$\because G$ 是有限群 $\therefore G$ 是有限集合. 定义 G 在 G 上的群作用为:

对 $\forall g \in G, \forall x \in G, \quad gx := gxg^{-1}$

则有: 对 $\forall x \in G, \quad 1_G x = 1_G x 1_G^{-1} = 1_G x 1_G = x.$

对 $\forall g_1, g_2 \in G, \forall x \in G$, 有:

$$g_1(g_2 x) = g_1(g_2 x g_2^{-1}) = g_1(g_2 x g_2^{-1}) g_1^{-1}$$

$$= (g_1 g_2) x (g_2^{-1} g_1^{-1}) = (g_1 g_2) x (g_1 g_2)^{-1} = (g_1 g_2) x$$

$\therefore$ 这是群 G 在 G 上的共轭作用.

已经证明了 $G^G = Z_G. \quad \because G$ 是 p -群, G 作用在非空有限集合 G 上

$$\therefore |G| \equiv |G^q| \pmod{p} \quad \therefore |G| \equiv |Z_G| \pmod{p}$$

$$\therefore |G| = p^m, m \in \mathbb{Z}_{\geq 1} \quad \therefore p \mid |G| \quad \therefore |G| \equiv 0 \pmod{p}$$

$$\therefore |Z_G| \equiv 0 \pmod{p} \quad \therefore \{1_G\} \subseteq Z_G \quad \therefore |Z_G| \in \mathbb{Z}_{\geq 1}$$

$$\therefore p \mid |Z_G| \quad \therefore |Z_G| = p^k, \text{ 其中 } k \in \mathbb{Z}_{\geq 1}$$

$$\therefore |Z_G| = p^k \geq 2 > 1 \quad \therefore \{1_G\} \subsetneq Z_G. \quad \square$$

Lemma: G 是一个任意的群, $n \in \mathbb{Z}_{\geq 2}$. 定义集合

$$G^n = \{(g_1, g_2, \dots, g_n) \mid \text{对 } \forall i=1, 2, \dots, n, \text{ 都有 } g_i \in G\}$$

$$G^{n-1} = \{(g_1, \dots, g_{n-1}) \mid \text{对 } \forall i=1, \dots, n-1, \text{ 都有 } g_i \in G\}.$$

$$X_n = \{(g_1, g_2, \dots, g_n) \in G^n \mid g_1 g_2 \cdots g_n = 1_G\}.$$

~~$$\text{定义映射 } \varphi: G^{n-1} \rightarrow X_n$$~~

~~$$(g_1, \dots, g_{n-1}) \mapsto (g_1, \dots, g_{n-1}, 1_G)$$~~

$$\text{定义映射 } \varphi: G^{n-1} \longrightarrow X_n$$

$$(g_1, \dots, g_{n-1}) \longmapsto (g_1, \dots, g_{n-1}, (g_1 \cdots g_{n-1})^{-1})$$

则有: 映射 φ 是双射. 因此 $|G^{n-1}| = |X_n|$.

Proof, 对 $\forall (g_1, \dots, g_{n-1}) \in G^{n-1}$, 有: $g_1 \in G, \dots, g_{n-1} \in G$

$$\therefore (g_1 \cdots g_{n-1})^{-1} \in G \quad \therefore g_1 \cdots g_{n-1} \cdot (g_1 \cdots g_{n-1})^{-1} = 1_G$$

$$\therefore \varphi((g_1, \dots, g_{n-1})) = (g_1, \dots, g_{n-1}, (g_1 \cdots g_{n-1})^{-1}) \in X_n.$$

$\therefore \varphi(G^{n-1}) \subseteq X_n$. 显然相等的自变量给出相等的映射值.

对 $\forall (g_1, \dots, g_{n-1}), (x_1, \dots, x_{n-1}) \in G^{n-1}$, 若

$\varphi((g_1, \dots, g_{n-1})) = \varphi((x_1, \dots, x_{n-1}))$, 则有

$$(g_1, \dots, g_{n-1}, (g_1 \cdots g_{n-1})^{-1}) = (x_1, \dots, x_{n-1}, (x_1 \cdots x_{n-1})^{-1})$$

$$\therefore g_1 = x_1, \dots, g_{n-1} = x_{n-1} \quad \therefore (g_1, \dots, g_{n-1}) = (x_1, \dots, x_{n-1})$$

$\therefore \varphi$ 是单射.

对 $\forall (g_1, \dots, g_{n-1}, g_n) \in X_n$, 有: $g_1 \cdots g_{n-1} g_n = |G|$

$$\therefore (g_1, \dots, g_{n-1}) \in G^{n-1}, \text{ 且 } g_n = (g_1 \cdots g_{n-1})^{-1}$$

$$\therefore \varphi((g_1, \dots, g_{n-1})) = (g_1, \dots, g_{n-1}, (g_1 \cdots g_{n-1})^{-1}) = (g_1, \dots, g_{n-1}, g_n)$$

$$\therefore \varphi \text{ 是满射.} \quad \therefore \varphi \text{ 是双射} \quad \therefore |G^{n-1}| = |X_n| \quad \square$$

定理(群论中的 Cauchy 定理) G 是有限群, p 是 $|G|$ 的素因数
则存在 $g \in G$ 使得 $\text{ord}(g) = p$

proof: $\because G$ 是有限群 $\therefore |G| \in \mathbb{Z}_{\geq 1}$ $\because p$ 是 $|G|$ 的素因数

$\therefore p$ 是素数且 $p \in \mathbb{Z}_{\geq 2}$ 且 $p \mid |G|$

$\therefore p \in \mathbb{Z}_{\geq 2} \quad \therefore (\mathbb{Z}/p\mathbb{Z}, +)$ 是循环群, 也是交换群.

$\because p \in \mathbb{Z}_{\geq 2} \quad \therefore$ 已经证明了 $\mathbb{Z}/p\mathbb{Z} = \{[0]_p, [1]_p, \dots, [p-1]_p\}$

$\because G$ 是有限群, $p \in \mathbb{Z}_{\geq 2}$

$\therefore G^p = \{(g_1, g_2, \dots, g_p) \mid \text{对 } \forall i=1, 2, \dots, p, \text{ 都有 } g_i \in G\}$

$G^{p-1} = \{(g_1, \dots, g_{p-1}) \mid \text{对 } \forall i=1, \dots, p-1, \text{ 都有 } g_i \in G\}$

$X = \{(g_1, g_2, \dots, g_p) \in G^p \mid g_1 g_2 \cdots g_p = 1_G\} \subseteq G^p$

已经证明了 $|X| = |G^{p-1}| = |G|^{p-1} \in \mathbb{Z}_{\geq 1}$

$\because p \in \mathbb{Z}_{\geq 2} \quad \therefore p-1 \in \mathbb{Z}_{\geq 1} \quad \therefore p \mid |G| \quad \therefore p \mid |G|^{p-1} \quad \therefore p \mid |X|$

定义映射 $\alpha: \mathbb{Z}/p\mathbb{Z} \times X \longrightarrow X$

$([k]_p, (g_1, g_2, \dots, g_p)) \longmapsto (g_{1+k}, g_{2+k}, \dots, g_{p+k})$

(下标运算在模 p 意义下进行)
(下标加法后取模 p 的最小正剩余)

任取 $\mathbb{Z}/p\mathbb{Z} \times X$ 中的一元: $([k]_p, (g_1, g_2, \dots, g_p))$ (其中 $k \in \mathbb{Z}$).

(其中 $(g_1, g_2, \dots, g_p) \in G^p$ 且 $g_1 g_2 \cdots g_p = 1_G$)

$\therefore \alpha([k]_p, (g_1, g_2, \dots, g_p)) = (g_{1+k}, g_{2+k}, \dots, g_{p+k}) \in G^p$

且 $g_{1+k} g_{2+k} \cdots g_{p+k} = g_1 g_2 \cdots g_p = 1_G \quad \text{---} (*)$

$\therefore \alpha([k]_p, (g_1, g_2, \dots, g_p)) \in X$

$\therefore \alpha(\mathbb{Z}/p\mathbb{Z} \times X) \subseteq X$

任取 $\mathbb{Z}/p\mathbb{Z} \times X$ 中的两个元素 $([k_1]_p, (g_1, g_2, \dots, g_p))$ 和

$([k_2]_p, (x_1, x_2, \dots, x_p))$, 若 $([k_1]_p, (g_1, g_2, \dots, g_p)) = ([k_2]_p, (x_1, x_2, \dots, x_p))$

则有: $k_1 \equiv k_2 \pmod{p}$ 且 $g_1 = x_1, g_2 = x_2, \dots, g_p = x_p$.

$$\therefore a\left(\left([k_1]_p, (g_1, g_2, \dots, g_p)\right)\right) = (g_{1+k_1}, g_{2+k_1}, \dots, g_{p+k_1})$$

~~$$= (x_{1+k_1}, x_{2+k_1}, \dots, x_{p+k_1})$$~~

$$(\because k_1 \equiv k_2 \pmod{p}, \alpha \equiv \alpha \pmod{p}) \quad \therefore \alpha + k_1 \equiv \alpha + k_2 \pmod{p}$$

$$= (x_{1+k_2}, x_{2+k_2}, \dots, x_{p+k_2}) = a\left(\left([k_2]_p, (x_1, x_2, \dots, x_p)\right)\right)$$

$\therefore a$ 是良定义的映射.

对 $\forall (g_1, g_2, \dots, g_p) \in X$, 有:

$$a\left([0]_p, (g_1, g_2, \dots, g_p)\right) = (g_{1+0}, g_{2+0}, \dots, g_{p+0}) = (g_1, g_2, \dots, g_p)$$

对 $\forall [k_1]_p, [k_2]_p \in \mathbb{Z}/p\mathbb{Z}, \forall (g_1, g_2, \dots, g_p) \in X$, 有:

~~$$a\left([k_1]_p, a\left([k_2]_p, (g_1, g_2, \dots, g_p)\right)\right)$$~~

~~$$= a\left([k_1]_p, (g_{1+k_2}, g_{2+k_2}, \dots, g_{p+k_2})\right)$$~~

~~$$= (g_{(1+k_2)+k_1}, g_{(2+k_2)+k_1}, \dots, g_{(p+k_2)+k_1})$$~~

~~$$= (g_{1+(k_2+k_1)}, g_{2+(k_2+k_1)}, \dots, g_{p+(k_2+k_1)})$$~~

~~$$= a\left([k_2+k_1]_p, (g_1, g_2, \dots, g_p)\right)$$~~

~~$$= a\left([k_1]_p + [k_2]_p, (g_1, g_2, \dots, g_p)\right) \quad \therefore a \text{ 是群 } (\mathbb{Z}/p\mathbb{Z}, +) \text{ 在集合 } X \text{ 上的群作用.}$$~~

$$\begin{aligned}
& a([k_1]_p, a([k_2]_p, (g_1, g_2, \dots, g_p))) \\
&= a([k_1]_p, (g_{1+k_2}, g_{2+k_2}, \dots, g_{p+k_2})) \\
&\quad (\text{记 } (y_1, y_2, \dots, y_p) = (g_{1+k_2}, g_{2+k_2}, \dots, g_{p+k_2})) \\
&= a([k_1]_p, (y_1, y_2, \dots, y_p)) \\
&= (y_{1+k_1}, y_{2+k_1}, \dots, y_{p+k_1}) \\
&= (g_{(1+k_1)+k_2}, g_{(2+k_1)+k_2}, \dots, g_{(p+k_1)+k_2}) \\
&= (g_{1+(k_1+k_2)}, g_{2+(k_1+k_2)}, \dots, g_{p+(k_1+k_2)}) \\
&= a([k_1+k_2]_p, (g_1, g_2, \dots, g_p)) \\
&= a([k_1]_p + [k_2]_p, (g_1, g_2, \dots, g_p))
\end{aligned}$$

$\therefore a$ 是群 $(\mathbb{Z}/p\mathbb{Z}, +)$ 在集合 X 上的群作用.

$$\therefore X^{\mathbb{Z}/p\mathbb{Z}} = \left\{ (g_1, g_2, \dots, g_p) \in X \mid \text{对 } \forall [k]_p \in \mathbb{Z}/p\mathbb{Z} \text{ 都有 } a([k]_p, (g_1, g_2, \dots, g_p)) = (g_1, g_2, \dots, g_p) \right\}$$

$$\text{下证 } X^{\mathbb{Z}/p\mathbb{Z}} = \{ (g, g, \dots, g) \in G^p \mid g^p = 1_G \}$$

任取 $X^{\mathbb{Z}/p\mathbb{Z}}$ 中的一元: $(g_1, g_2, \dots, g_p)$, 有: $(g_1, g_2, \dots, g_p) \in X$

$\therefore g_1 \in G, g_2 \in G, \dots, g_p \in G$, 且 $g_1 g_2 \cdots g_p = 1_G$.

对于 $[1]_p \in \mathbb{Z}/p\mathbb{Z}$, 有: $\alpha([1]_p, (g_1, g_2, \dots, g_p)) = (g_1, g_2, \dots, g_p)$

$$\therefore (g_{p+1}, g_{p+2}, \dots, g_{p+p}) = (g_1, g_2, \dots, g_p)$$

$$\therefore (g_2, g_3, \dots, g_{p+1}) = (g_1, g_2, \dots, g_p)$$

$$\therefore g_1 = g_2, g_2 = g_3, \dots, g_{p-1} = g_p, g_p = g_{p+1} = g_1$$

$$\therefore g_1 = g_2 = \dots = g_p \text{ 且 } g_1 g_2 \dots g_p = 1_G$$

$$\therefore (g_1, g_2, \dots, g_p) \in \text{右边}. \quad \therefore \text{左边} \subseteq \text{右边}.$$

任取右边中的一元: $(g, g, \dots, g) \in G^p$ 且 $g^p = 1_G$, 有:

$$(g, g, \dots, g) \in X. \quad \text{对 } \forall [k]_p \in \mathbb{Z}/p\mathbb{Z}, \text{ 都有}$$

$$\alpha([k]_p, (g, g, \dots, g)) = (g, g, \dots, g)$$

$$\therefore (g, g, \dots, g) \in X^{\mathbb{Z}/p\mathbb{Z}} \quad \therefore \text{右边} \subseteq \text{左边} \quad \therefore \text{左边} = \text{右边}.$$

考虑 $(1_G, 1_G, \dots, 1_G)$ (p 个 1_G 分量).

$$\therefore (1_G, 1_G, \dots, 1_G) \in G^p \text{ 且 } 1_G \cdot 1_G \cdot \dots \cdot 1_G = 1_G$$

$$\therefore (1_G, 1_G, \dots, 1_G) \in X. \quad (\text{因此 } X \text{ 是非空有限集合})$$

$$\text{对 } \forall [k]_p \in \mathbb{Z}/p\mathbb{Z} \text{ 都有 } \alpha([k]_p, (1_G, 1_G, \dots, 1_G)) = (1_G, 1_G, \dots, 1_G)$$

$$\therefore (1_G, 1_G, \dots, 1_G) \in X^{\mathbb{Z}/p\mathbb{Z}} \quad \therefore |X^{\mathbb{Z}/p\mathbb{Z}}| \geq 1.$$

$$\therefore |\mathbb{Z}/p\mathbb{Z}| = p \quad \therefore (\mathbb{Z}/p\mathbb{Z}, +) \text{ 是 } p\text{-群}. \quad \therefore |X| \equiv |X^{\mathbb{Z}/p\mathbb{Z}}| \pmod{p}$$

$$\because p \mid |X| \quad \therefore |X| \equiv 0 \pmod{p} \quad \therefore |X^{\mathbb{Z}/p\mathbb{Z}}| \equiv 0 \pmod{p}$$

$$\therefore p \mid |X^{\mathbb{Z}/p\mathbb{Z}}| \quad \therefore X^{\mathbb{Z}/p\mathbb{Z}} \subseteq X, \quad X \text{ 是非空有限集合}$$

$$\therefore X^{\mathbb{Z}/p\mathbb{Z}} \text{ 是非空有限集合.} \quad \therefore |X^{\mathbb{Z}/p\mathbb{Z}}| \in \mathbb{Z}_{\geq 1}$$

$$\therefore |X^{\mathbb{Z}/p\mathbb{Z}}| \geq p \geq 2 \quad \therefore X^{\mathbb{Z}/p\mathbb{Z}} \text{ 中除了 } (1_g, 1_g, \dots, 1_g) \text{ 之外, 还有别的} \\ (p \text{ 个 } 1_g \text{ 分量})$$

$$\text{元素. 设 } (\lambda, \lambda, \dots, \lambda) \in X^{\mathbb{Z}/p\mathbb{Z}}$$

$$(\text{其中 } (\lambda, \lambda, \dots, \lambda) \in G^p \text{ 且 } \lambda^p = 1_g \text{ 且 } (\lambda, \lambda, \dots, \lambda) \neq (1_g, 1_g, \dots, 1_g))$$

$$\therefore \lambda \in G, \lambda \neq 1_g, \lambda^p = 1_g, p \in \mathbb{Z}_{\geq 2} \text{ 且 } p \text{ 是素数}$$

$$\because G \text{ 是有限群, } \lambda \in G \quad \therefore \text{ord}(\lambda) \mid |G| \quad \therefore \text{ord}(\lambda) \in \mathbb{Z}_{\geq 1}$$

$$\because G \text{ 是有限群, } \lambda \in G, \text{ord}(\lambda) \in \mathbb{Z}_{\geq 1}, p \in \mathbb{Z}_{\geq 2}, \lambda^p = 1_g$$

$$\therefore \text{ord}(\lambda) \mid p \quad \therefore \text{ord}(\lambda) = 1 \text{ 或 } p.$$

$$\text{假设 } \text{ord}(\lambda) = 1 \quad \text{则 } \lambda = 1_g. \text{ 矛盾.} \quad \therefore \text{ord}(\lambda) = p.$$

$$\therefore \lambda \in G \text{ 且 } \text{ord}(\lambda) = p. \quad \square$$

Remark: (*) 处的证明需要更多的说明.

$$\text{当 } k=0 \text{ 时, } g_{1+k} g_{2+k} \cdots g_{p+k} = g_{1+0} g_{2+0} \cdots g_{p+0} = g_1 g_2 \cdots g_p = 1_g.$$

$$\text{当 } k=1 \text{ 时, } g_{1+k} g_{2+k} \cdots g_{p+k} = g_{1+1} g_{2+1} \cdots g_{p+1} = g_2 g_3 \cdots g_p g_1 = g_1^{-1} g_1 = 1_g.$$

$$\text{当 } k=2 \text{ 时, } g_{1+k} g_{2+k} \cdots g_{p+k} = g_{1+2} g_{2+2} \cdots g_{p+2} = g_3 g_4 \cdots g_p g_1 g_2 = (g_1 g_2)^{-1} (g_1 g_2) = 1_g$$

$$\text{当 } k=3 \text{ 时, } g_{1+k} g_{2+k} \cdots g_{p+k} = g_{1+3} g_{2+3} \cdots g_{p+3} = g_4 g_5 \cdots g_p g_1 g_2 g_3 \\ = (g_1 g_2 g_3)^{-1} (g_1 g_2 g_3) = I_q.$$

将上述过程继续进行下去, 可以证明 对 $\forall k \in \mathbb{Z}$ 都有

$$g_{1+k} g_{2+k} \cdots g_{p+k} = I_q.$$